UGB Toppers' Test-3

ISI Preparation · Mock Paper

Time: 2 hours | Maximum marks: 80
Number of questions: 8 | Marks per question: 10

Instructions. Attempt all eight problems. Each problem carries equal weight; partial credit will be awarded for substantial progress. Write your solutions clearly and justify every step. Calculators and electronic devices are not permitted. Standard notation is used throughout: $\mathbb{R}$ denotes the real numbers, $\mathbb{C}$ the complex numbers, $\binom{n}{k}$ the usual binomial coefficient, and $[XYZW]$ the area of quadrilateral $XYZW$.

Problem 1. Prove that there exists an infinite sequence of positive integers $\{a_n\}_{n\geq 1}$ such that $a_1^2 + a_2^2 + \cdots + a_n^2$ is a perfect square for every $n \geq 1$.

Problem 2. Let $z_1, z_2, \ldots, z_n$ be non-zero complex numbers, all lying strictly on one side of some straight line through the origin of the complex plane. Prove that

$$\sum_{k=1}^{n} z_k \neq 0 \qquad \text{and} \qquad \sum_{k=1}^{n} \frac{1}{z_k} \neq 0.$$

Problem 3. Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Suppose $f(a) = f(b) = 0$ and f is not identically zero on $[a, b]$. Prove that for every positive integer n there exists a point $c \in (a, b)$ such that

$$(b - c)\, f'(c) \;=\; n\, f(c).$$

Problem 4. Let $f : [0, 2\pi] \to \mathbb{R}$ be a continuous function such that

$$\int_0^{2\pi} f(x)\, dx \;=\; \int_0^{2\pi} f(x) \cos x\, dx \;=\; \int_0^{2\pi} f(x) \sin x\, dx \;=\; 0.$$

Prove that f has at least three distinct zeros in the interval $(0, 2\pi)$.

Problem 5. Let $ABCD$ be a convex quadrilateral inscribed in a parabola. The tangents to the parabola at A, B, C, D meet pairwise at P, Q, R, S, where

$$P = \tan_A \cap \tan_B, \quad Q = \tan_B \cap \tan_C, \quad R = \tan_C \cap \tan_D, \quad S = \tan_D \cap \tan_A.$$

Prove that $[ABCD] = 2 \cdot [PQRS]$.

Problem 6. Let ABC be a triangle with angles A, B, C. Prove that

$$\frac{\cos A}{\sin^3 A} + \frac{\cos B}{\sin^3 B} + \frac{\cos C}{\sin^3 C} \ \geq \ \frac{3}{2 \sin A \sin B \sin C}.$$

Problem 7. Let $f(x)$ be a real-coefficient polynomial of degree at most $2n$ such that $|f(k)| \leq 1$ for every integer k with $-n \leq k \leq n$. Prove that

$$|f(x)| \ \leq \ 2^{2n} \qquad \text{for every real } x \text{ with } \ -n \leq x \leq n.$$

Problem 8. Let p be an odd prime. Prove that

$$\sum_{j=0}^{p} \binom{p}{j} \binom{p+j}{j} \ \equiv \ 2^p + 1 \pmod{p^2}.$$

End of paper